ANALISI 1 : 02/09/2025

a) DEF (MAX MIN LOCALE): $I \subseteq \mathbb{R}$, $f: I \to \mathbb{R}$, $x_0 \in I$. SI DICE CHE x_0 È UN PUNTO DI MASSIMO (MINIMO) LOCALE PER f SE:

$$\exists\, U \text{ INTORNO DI } x_0 \text{ t.c. } f(x_0) \geq f(x) \quad \forall x \in I \cap U \setminus \{x_0\}$$

b) TEO (DEGLI ZERI): $f: [a,b] \to \mathbb{R}$, $f \in C^0([a,b])$. SUPPONIAMO CHE $f(a) \cdot f(b) < 0$, OVVERO HANNO SEGNI DISCORDI, ALLORA:

$$\exists\, \hat{x} \in (a,b) \text{ t.c. } f(\hat{x}) = 0.$$

c) DIMOSTRARE: $(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0)$

$$(fg)'(x_0) = \lim_{x \to x_0} \frac{f(x)g(x) - f(x_0)g(x_0)}{x - x_0} =$$

$$= \lim_{x \to x_0} \frac{f(x)g(x) + f(x_0)g(x) - f(x_0)g(x) - f(x_0)g(x_0)}{x - x_0} =$$

$$= \lim_{x \to x_0} \underbrace{g(x)}_{g(x_0)} \Bigg(\underbrace{\frac{f(x) - f(x_0)}{x - x_0}}_{f'(x_0)} \Bigg) + f(x_0) \Bigg(\underbrace{\frac{g(x) - g(x_0)}{x - x_0}}_{g'(x_0)} \Bigg) = f'(x_0)g(x) + f(x_0)g'(x_0)$$

[redacted]

↳ VISTO $g(x)$ È DERIVABILE IN x_0 ALLORA È CONTINUO IN x_0

ESERCIZIO 2: $f = \dfrac{\sqrt{\log(x)}}{x}$

DOMINIO: $\log(x) \geq 0$ se $\log(x) \geq \log(1)$ $\quad x \geq 1$

$x > 0 \quad x \neq 0 \quad D = [1, +\infty)$

SIMMETRIE: NESSUNA (IL DOMINIO NON LO PERMETTE)

SEGNO: $f(x) \geq 0$ se $\dfrac{\sqrt{\log(x)}}{x} \geq 0 \quad \sqrt{\log(x)} \geq 0 \quad x \geq 1$

$f(x) \geq 0 \ \forall x \in D \quad f(x) < 0$ MAI

LIMITI: $\displaystyle\lim_{x \to +\infty} \frac{\sqrt{\log(x)}}{x} = 0$ ASINTOTO ORIZZONTALE $y = 0$

NO ASINTOTO OBLIQUO

DERIVATA PRIMA: $f'(x) = \left(\dfrac{\sqrt{\log(x)}}{x}\right)' = \dfrac{\frac{1}{2\sqrt{\log(x)}} \cdot \frac{1}{\cancel{x}} \cdot \cancel{x} - \sqrt{\log(x)}}{x^2} =$

$$= \frac{\frac{1}{2}(\log(x))^{-\frac{1}{2}} - (\log(x))^{+\frac{1}{2}}}{x^2}$$

MONOTONIA: $f'(x) \ge 0$ se $\dfrac{\frac{1}{2}(\log(x))^{-\frac{1}{2}} - (\log(x))^{+\frac{1}{2}}}{x^2} \ge 0$ $\quad x^2 \to >0 \ \forall x \in D$

$$(\log(x))^{-\frac{1}{2}} \left(\frac{1}{2} - (\log(x))^{1}\right) \ge 0$$

$$(\log(x))^{-\frac{1}{2}} > 0 \qquad \frac{1}{\sqrt{\log(x)}} > 0 \quad \forall x \in D$$

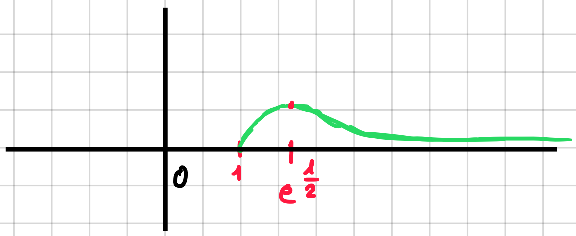
$\left(\frac{1}{2} - \log(x)\right) \ge 0$ se $\log(x) \le \frac{1}{2}$ $\qquad x \le e^{\frac{1}{2}} \Rightarrow 1 < x \le e^{\frac{1}{2}}$

$f'(x) > 0 \qquad 1 < x < e^{\frac{1}{2}}$ CRESCE

$f'(x) = 0 \qquad x = e^{\frac{1}{2}}$ ESTREMO LOCALE (MASSIMO ASSOLUTO)

$f'(x) < 0 \qquad x > e^{\frac{1}{2}}$

GRAFICO:



ESERCIZIO 3: $\lim\limits_{x \to 0} \dfrac{1-\cos(5x)}{x\tan(3x)}$ $\qquad 1-\cos(5x) \sim \frac{1}{2}(5x)^2$ PER $x \to 0$

$\tan(3x) \sim 3x$ PER $x \to 0$

$$\lim_{x \to 0} \frac{\frac{1}{2}\,25\cancel{x^2}}{3\cancel{x^2}} = \frac{25}{6}$$

$$\lim_{x \to +\infty} \sqrt{x^2+6x+1} - x = \lim_{x \to +\infty} \sqrt{x^2+6x+1} - x \cdot \left(\frac{\sqrt{x^2+6x+1} + x}{\sqrt{x^2+6x+1} + x}\right) =$$

$$= \lim_{x \to +\infty} \frac{\cancel{x^2}+6x+1-\cancel{x^2}}{\sqrt{x^2+6x+1} + x} = \lim_{x \to +\infty} \frac{\cancel{x}\left(6 + \frac{1}{x}\right)}{\cancel{x}\left(\sqrt{1+\underset{\downarrow 0}{\frac{6}{x}}+\underset{\downarrow 0}{\frac{1}{x^2}}} + 1\right)} = \frac{6}{2} = 3$$

ESERCIZIO 4:

a) $\displaystyle\int x^2 \log(x)\,dx = \frac{1}{3}x^3\log(x) - \int \frac{1}{3}x^{\cancel{3}\,2} \cdot \frac{1}{\cancel{x}}\,dx =$

$$= \frac{1}{3}x^3\log(x) - \frac{1}{3}\int x^2\,dx = \frac{1}{3}x^3\log(x) - \frac{1}{9}x^3 + c$$

D	$\int$
$\frac{1}{x} \leftarrow$	$\log(x)$
$x^2 \rightarrow$	$\frac{x^3}{3}$

b) $\displaystyle\int \frac{x^2}{x^2+4}\,dx$

$$\begin{array}{rr|l} x^2 & & x^2+4 \\ \hline -x^2 & -4 & 1 \\ \hline / & -4 & \end{array}$$

$$\int\left(1 - \frac{4}{x^2+4}\right)dx = \int 1\,dx - 4\int\frac{1}{x^2+4}\,dx = \int 1\,dx - 4\cdot\frac{1}{4}\int\frac{1}{\left(\frac{x}{2}\right)^2+1}\,dx =$$

$$\int 1\,dx - 2\int\frac{\frac{1}{2}}{\left(\frac{x}{2}\right)^2+1}\,dx = x - 2\arctan\left(\frac{x}{2}\right) + c$$